LAB-13: Neural Network Implementation – Hyperparameter Setting and Regularization

Course- B. Tech.	**Type-** Core
Course Code- CSET301	**Course Name-** Artificial Intelligence and Machine Learning
Year- 2025	**Semester-** Odd
Date-	**Batch-** 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
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Objectives

- Understand the role of hyperparameters in training a neural network, including learning rate, hidden layer sizes, and regularization (L2 penalty).

- Gain practical experience in adjusting these hyperparameters and observing their effects on training accuracy, validation accuracy, and test accuracy.

- Analyze how smaller or larger hidden layers affect the model's ability to generalize.

- Experiment with different values of the learning rate to see how it influences convergence speed and stability.

- Explore the role of L2 regularization (alpha) in controlling overfitting by penalizing large weights.

- Perform multiple experiments with different hyperparameter combinations and systematically compare results.

- Record your observations in a structured table and provide conclusions about which hyperparameters work best for the given dataset.

Steps to Solve

- Load the Breast Cancer dataset from scikit-learn.

- Split the dataset into training and testing sets.

- Standardize the data using StandardScaler to ensure better convergence of the neural network.

- Choose at least 4-5 combinations of hyperparameters (hidden layers, alpha for regularization, and learning rate).

- Train the model with MLPClassifier for each combination.

- Record the training and test accuracy for each experiment.

- Analyze results and write a short conclusion about the impact of hyperparameter choices.